

DR. D. Y. PATIL INSTITUTE OF ENGINEERING, MANAGEMENT AND RESEARCH
Akurdi, Pune-411044

Department-Artificial Intelligence and Data Science

Laboratory Journal | Subject: Computer Laboratory – IV

AY:2025-26

Name: Asad Alwase

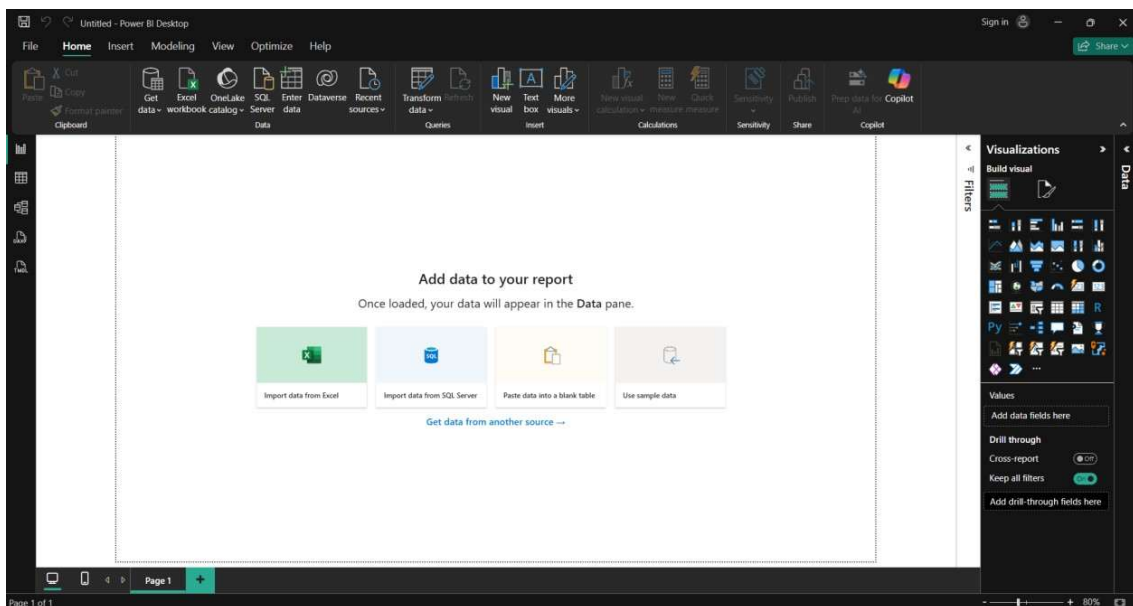
Class: B.E. (A)

Roll No: 22542

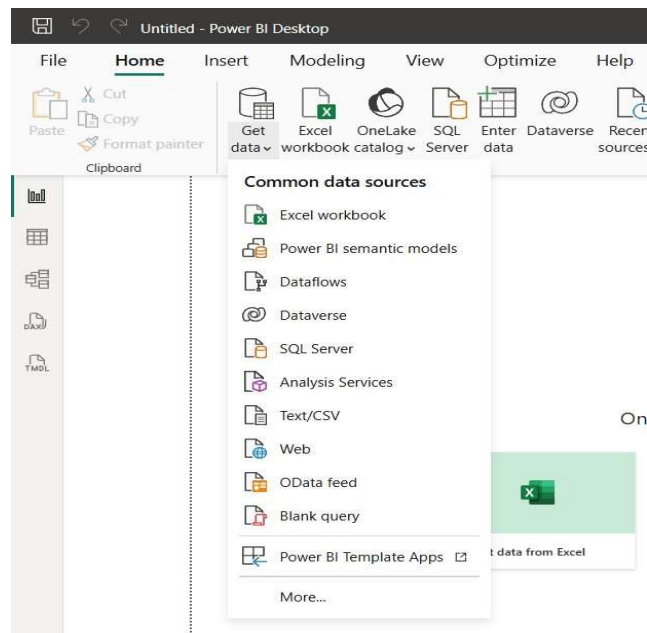
Lab Practical 7

Import Data from different Sources such as (Excel, Sql Server, Oracle etc.) and load in targeted system.

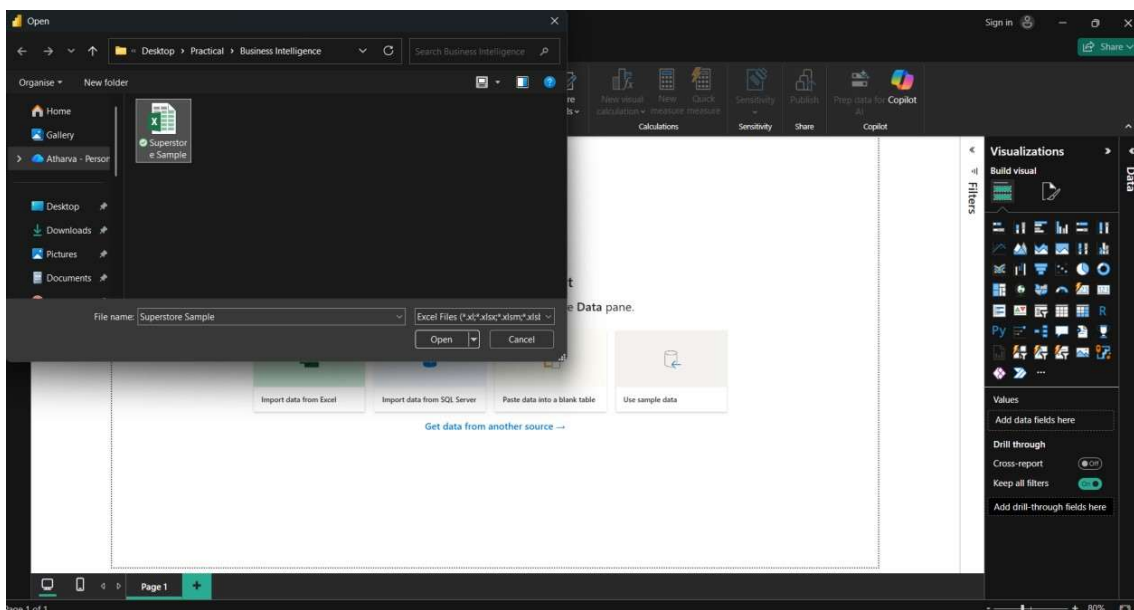
1) Launch Power BI Desktop.



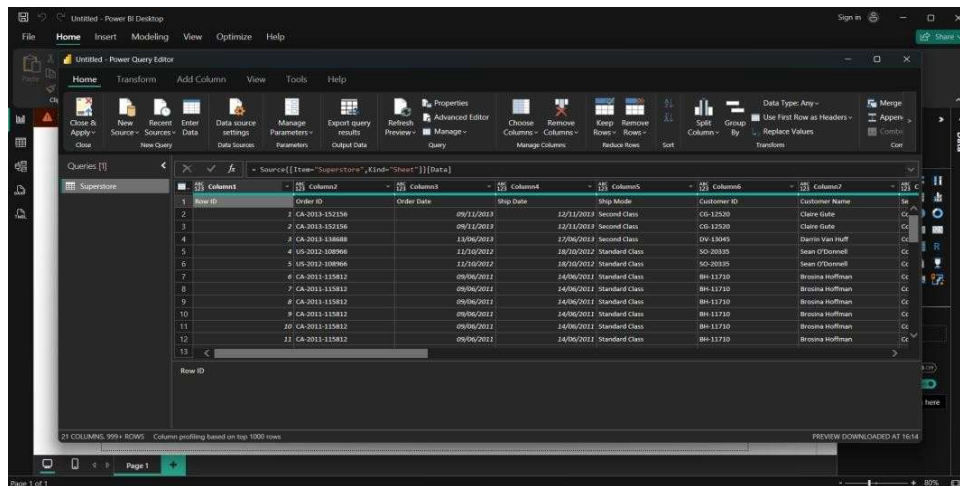
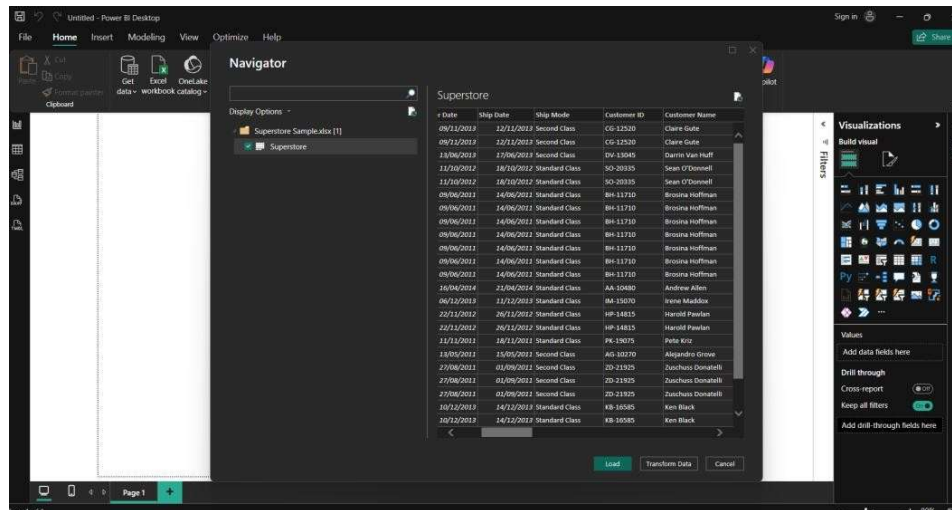
2) From the Home ribbon, select Get Data.



3) In the Open File dialog box, select the Products.xlsx file.



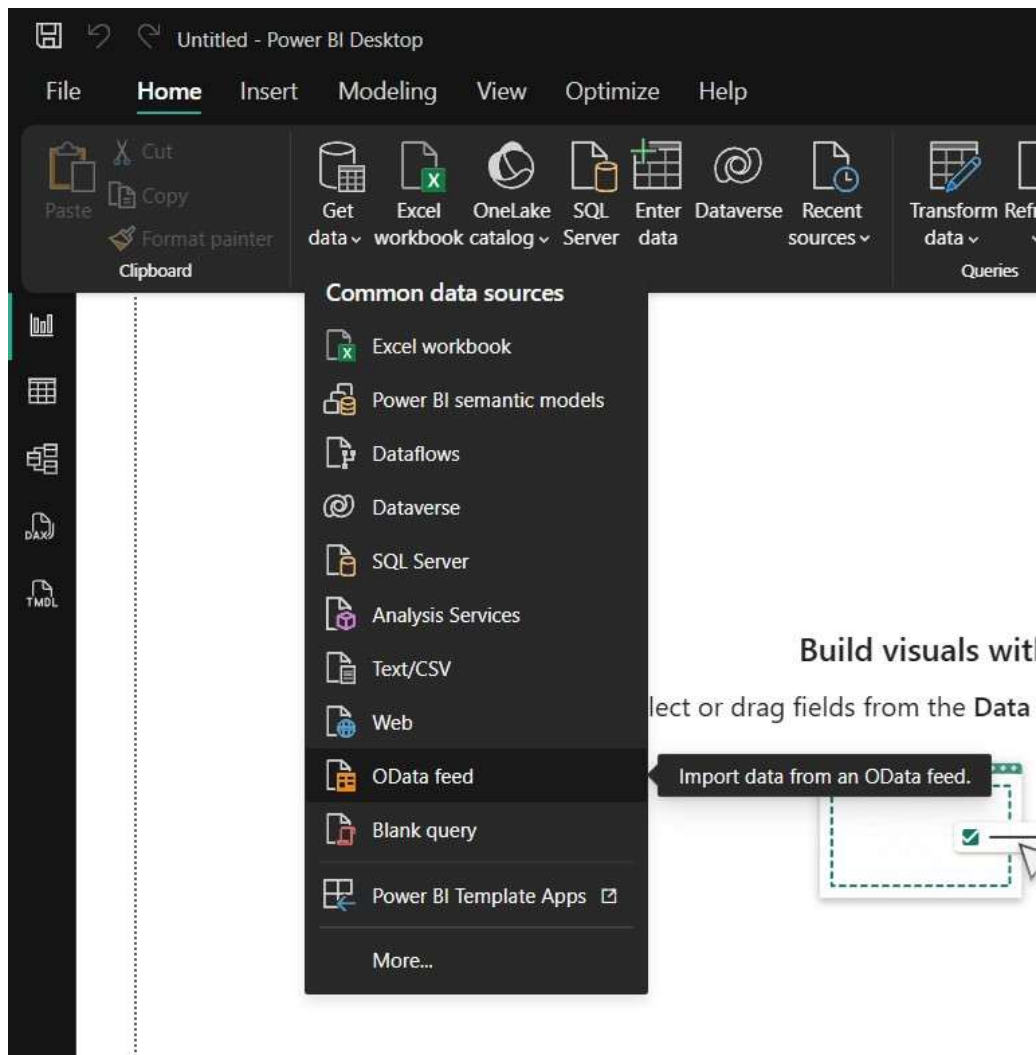
4) In the Navigator pane, select the Products table and then select Edit.



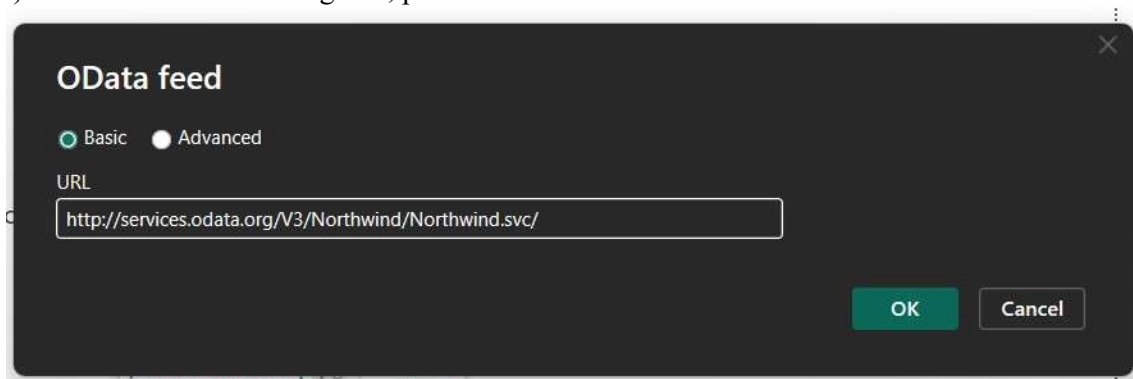
Importing Data from OData Feed

(<http://services.odata.org/V3/Northwind/Northwind.svc/>)

- 1) From the Home ribbon tab in Query Editor, select Get Data. Browse to the OData Feed data source.



2) In the OData Feed dialog box, paste the URL for the Northwind OData feed.



3) In the Navigator pane, select the Orders table, and then select Edit

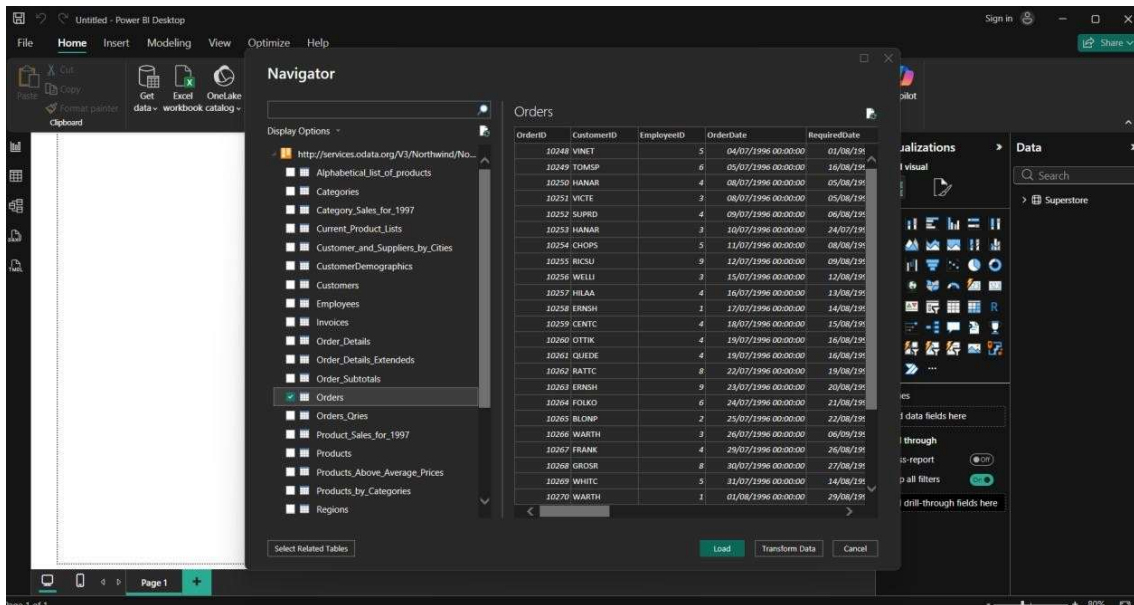


Table: Orders (830 rows)

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	ShipVia	Freight	ShipName	ShipAddress	ShipCity	ShipRegion	ShipPost
10249	TOMSP	6	05/07/1996 00:00:00	16/08/1996 00:00:00	10/07/1996 00:00:00	1	71.67	Toms Spezialitäten	Luisenstr. 48	Münster		44087
10260	OTTIK	4	19/07/1996 00:00:00	16/08/1996 00:00:00	29/07/1996 00:00:00	1	55.09	Ottiles Käseladen	Mehrhmeistr. 369	Köln		50739
10267	FRANK	4	29/07/1996 00:00:00	26/08/1996 00:00:00	06/08/1996 00:00:00	1	208.58	Frankenversand	Berliner Platz 43	München		80805
10273	QUICK	3	05/08/1996 00:00:00	02/09/1996 00:00:00	12/08/1996 00:00:00	3	76.07	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10277	MORGK	2	09/08/1996 00:00:00	06/09/1996 00:00:00	13/08/1996 00:00:00	3	125.77	Morgenstern Gesundkost	Heerstr. 22	Leipzig		04179
10279	LEHMS	8	13/08/1996 00:00:00	10/09/1996 00:00:00	16/08/1996 00:00:00	2	25.83	Lehmanns Marktstand	Magazinweg 7	Frankfurt a.M.		60528
10284	LEHMS	4	19/08/1996 00:00:00	16/09/1996 00:00:00	27/08/1996 00:00:00	1	76.56	Lehmanns Marktstand	Magazinweg 7	Frankfurt a.M.		60528
10285	QUICK	1	20/08/1996 00:00:00	17/09/1996 00:00:00	26/08/1996 00:00:00	2	76.83	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10286	QUICK	8	21/08/1996 00:00:00	18/09/1996 00:00:00	30/08/1996 00:00:00	3	229.24	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10301	WANIK	8	09/09/1996 00:00:00	07/10/1996 00:00:00	17/09/1996 00:00:00	2	45.08	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10312	WANIK	2	23/09/1996 00:00:00	21/10/1996 00:00:00	03/10/1996 00:00:00	2	40.26	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10313	QUICK	2	24/09/1996 00:00:00	22/10/1996 00:00:00	04/10/1996 00:00:00	2	1.96	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10323	KDENE	4	07/10/1996 00:00:00	04/11/1996 00:00:00	14/10/1996 00:00:00	1	4.88	Königlich Essen	Maubelstr. 90	Brandenburg		14776
10325	KDENE	1	08/10/1996 00:00:00	23/10/1996 00:00:00	14/10/1996 00:00:00	3	64.86	Königlich Essen	Maubelstr. 90	Brandenburg		14776
10337	FRANK	4	24/10/1996 00:00:00	21/11/1996 00:00:00	29/10/1996 00:00:00	3	108.26	Frankenversand	Berliner Platz 43	München		80805
10342	FRANK	4	30/10/1996 00:00:00	13/11/1996 00:00:00	04/11/1996 00:00:00	2	54.83	Frankenversand	Berliner Platz 43	München		80805
10343	LEHMS	4	31/10/1996 00:00:00	28/11/1996 00:00:00	06/11/1996 00:00:00	1	110.37	Lehmanns Marktstand	Magazinweg 7	Frankfurt a.M.		60528
10345	QUICK	2	04/11/1996 00:00:00	02/12/1996 00:00:00	11/11/1996 00:00:00	2	249.06	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10348	WANIK	4	07/11/1996 00:00:00	05/12/1996 00:00:00	15/11/1996 00:00:00	2	0.78	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10356	WANIK	6	18/11/1996 00:00:00	16/12/1996 00:00:00	27/11/1996 00:00:00	2	36.71	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10361	QUICK	1	22/11/1996 00:00:00	20/12/1996 00:00:00	03/12/1996 00:00:00	2	183.17	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10363	DRACD	4	26/11/1996 00:00:00	24/12/1996 00:00:00	04/12/1996 00:00:00	3	30.54	Drachenblut Delikatessen	Waisenweg 21	Aachen		52066
10391	DRACD	3	23/12/1996 00:00:00	20/01/1997 00:00:00	31/12/1996 00:00:00	3	5.45	Drachenblut Delikatessen	Waisenweg 21	Aachen		52066
10396	FRANK	1	27/12/1996 00:00:00	10/01/1997 00:00:00	06/01/1997 00:00:00	3	135.35	Frankenversand	Berliner Platz 43	München		80805
10407	OTTIK	2	07/01/1997 00:00:00	04/02/1997 00:00:00	30/01/1997 00:00:00	2	91.48	Ottiles Käseladen	Mehrhmeistr. 369	Köln		50739
10418	QUICK	4	17/01/1997 00:00:00	14/02/1997 00:00:00	24/01/1997 00:00:00	1	77.55	QUICK-Stop	Taucherstraße 10	Cunewalde		01307

Sign: _____.

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Lab Practical 8

Data Visualization from Extraction Transformation and Loading (ETL) Process.

Step 1: Check Data (Extraction)

```
import pandas as pd
import matplotlib.pyplot as plt

# EXTRACT - Load raw data from CSV
df = pd.read_csv("Superstore.csv", encoding='latin1')

print("Shape of data:", df.shape)
print("\nFirst 5 rows:")
df.head()
```

✓ 5.2s

Shape of data: (9994, 21)

Step 2: Transform Check-missing values

```
# Check missing values
print("Missing values:\n", df.isnull().sum())

# Drop rows with null values
df = df.dropna()

# Select only required columns
df = df[['Category', 'Sub-Category', 'Sales', 'Profit', 'Quantity']]

✓ 0.0s
```

Missing values:

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0
dtype: int64	

Remove Duplicates if any


```
df = df.drop_duplicates()

print("\nCleaned Data Shape:", df.shape)
print(df.head())
```

✓ 0.3s

Cleaned Data Shape: (7729, 5)

	Category	Sub-Category	Sales	Profit	Quantity
0	Furniture	Bookcases	261.9600	41.9136	2
1	Furniture	Chairs	731.9400	219.5820	3
2	Office Supplies	Labels	14.6200	6.8714	2
3	Furniture	Tables	957.5775	-383.0310	5
4	Office Supplies	Storage	22.3680	2.5164	2

Step 3: Load

```
# LOAD - Save cleaned data to new CSV
df.to_csv("Superstore_cleaned.csv", index=False)

print("Cleaned data loaded/saved successfully!")
```

✓ 0.0s

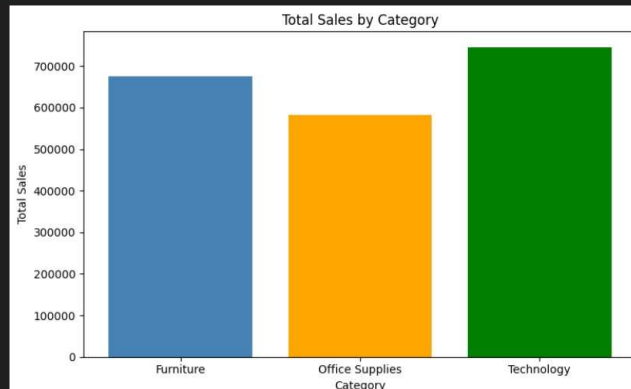
Cleaned data loaded/saved successfully!

Step 4: Visualization

```
category_sales = df.groupby('Category')['Sales'].sum()

plt.figure(figsize=(8,5))
plt.bar(category_sales.index, category_sales.values, color=['steelblue','orange','green'])
plt.title('Total Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.tight_layout()
plt.show()
```

✓ 0.3s





Sign: _____.

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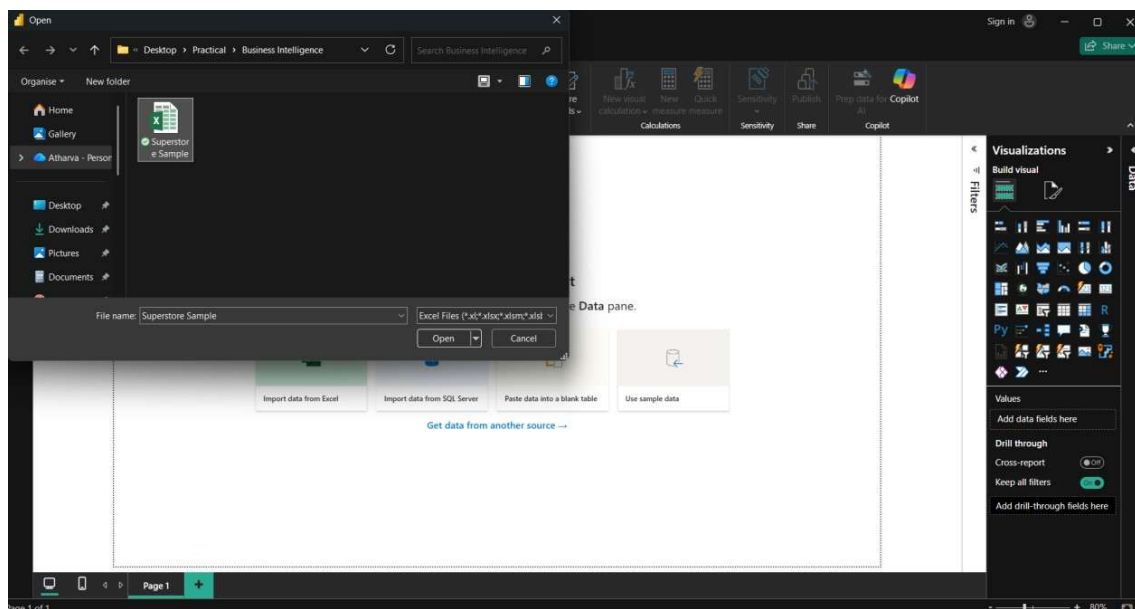
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Lab Practical 9

Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

STEP 1: Extraction (Load Data)



Navigator

Display Options

- Superstore Sample.xlsx (1)
- Superstore

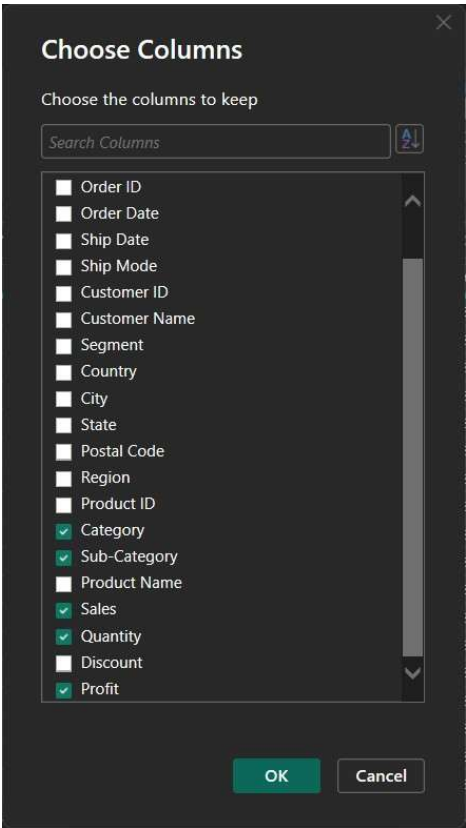
Superstore

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Cu
1	CA-2013-152156	09/11/2013	12/11/2013	Second Class	
2	CA-2013-152156	09/11/2013	12/11/2013	Second Class	
3	CA-2013-138688	13/06/2013	17/06/2013	Second Class	
4	US-2012-108966	11/10/2012	18/10/2012	Standard Class	
5	US-2012-108966	11/10/2012	18/10/2012	Standard Class	
6	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
7	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
8	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
9	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
10	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
11	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
12	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
13	CA-2014-114412	16/04/2014	21/04/2014	Standard Class	
14	CA-2013-161389	06/12/2013	11/12/2013	Standard Class	
15	US-2012-118983	22/11/2012	26/11/2012	Standard Class	
16	US-2012-118983	22/11/2012	26/11/2012	Standard Class	
17	CA-2011-105893	11/11/2011	18/11/2011	Standard Class	
18	CA-2011-167164	13/05/2011	15/05/2011	Second Class	
19	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
20	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
21	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
22	CA-2013-137330	10/12/2013	14/12/2013	Standard Class	
23	CA-2013-137330	10/12/2013	14/12/2013	Standard Class	

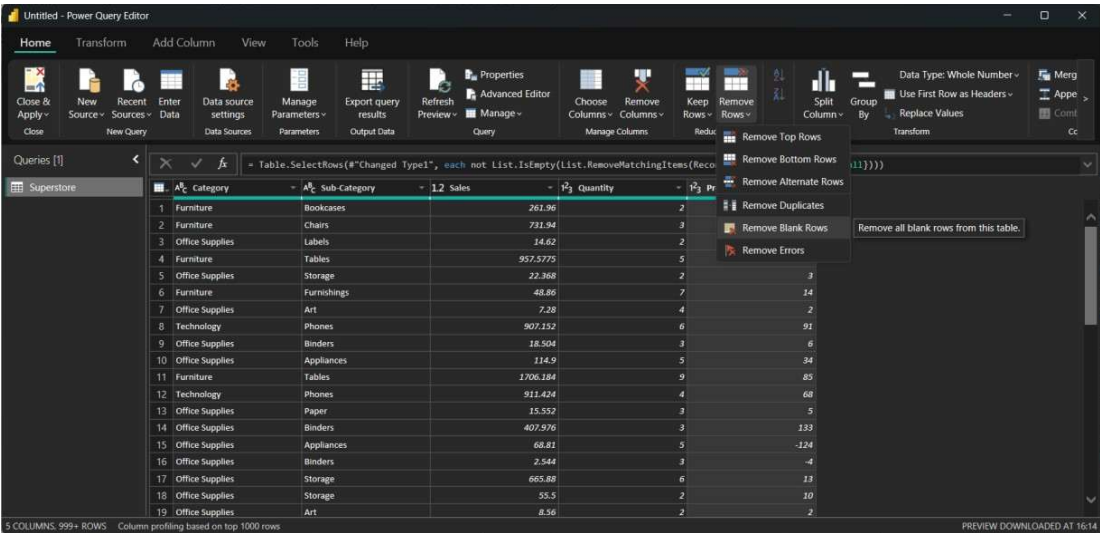
Load Transform Data Cancel

STEP 2: Transformation

(A) Remove Unnecessary Columns (Select the ones you want to keep)



(B) Handle Missing Values



(C) Changing the Data Type:

Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By

Queries [1] Superstore

fx = Table.TransformColumnTypes(#"Removed Other Columns",{"Sales", type number})

Category	Sub-Category	Sales	Quantity	Profit
Furniture	Bookcases	261.96	2	41.9136
Furniture	Chairs	731.94	3	219.582
Office Supplies	Labels	14.62	2	6.8714
Furniture	Tables	957.5775	5	383.021
Office Supplies	Storage	22.368	2	2.5164
Furniture	Furnishings	48.86	7	14.1694
Office Supplies	Art	7.28	4	1.9656
Technology	Phones	907.152	6	90.7152
Office Supplies	Binders	18.504	3	5.7825
Office Supplies	Appliances	114.9	5	34.47
Furniture	Tables	1706.184	9	85.3092
Technology	Phones	911.424	4	68.3568
Office Supplies	Paper	15.552	3	5.4432
Office Supplies	Binders	407.976	3	132.5922
Office Supplies	Appliances	68.81	5	123.858
Office Supplies	Binders	2.544	3	-3.816
Office Supplies	Storage	665.88	6	13.3176
Office Supplies	Storage	55.5	2	9.99
Office Supplies	Art	8.56	2	2.4824

5 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 16:14

(D) Create NEW Column

Untitled - Power Query Editor

Home Transform Add Column View Tools Help

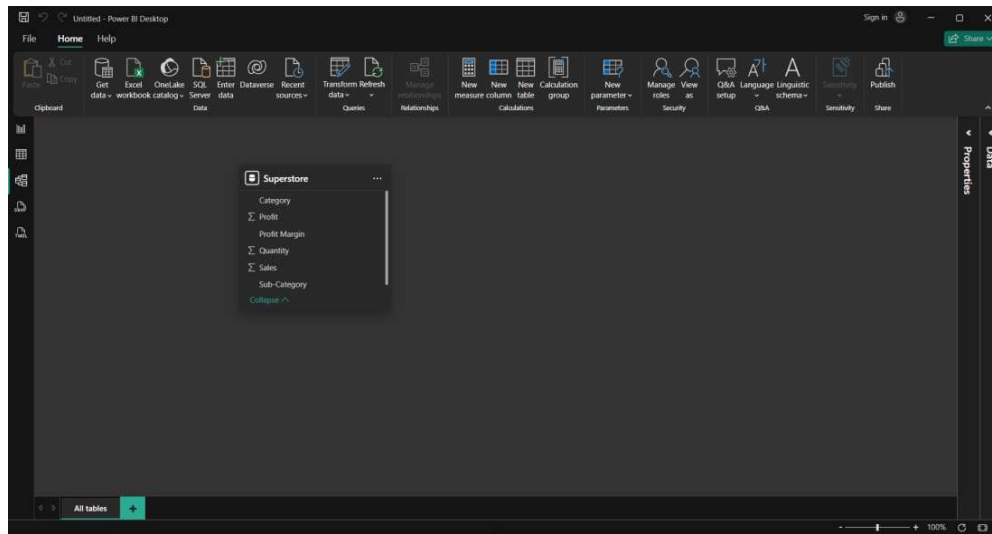
Column From Examples Custom Column Invoke Custom Function Duplicate Column Conditional Column Index Column Merge Columns Extract Format Parse Statistics Standard Scientific Trigonometry Rounding Date Time Duration

Create a new column in this table, based on a custom formula.

fx = Table.SelectRows(#"Changed Type1", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))

Category	Sub-Category	Sales	Quantity	Profit
Furniture	Bookcases	261.96	2	42
Furniture	Chairs	731.94	3	220
Office Supplies	Labels	14.62	2	7
Furniture	Tables	957.5775	5	-383
Office Supplies	Storage	22.368	2	3
Furniture	Furnishings	48.86	7	14
Office Supplies	Art	7.28	4	2
Technology	Phones	907.152	6	91
Office Supplies	Binders	18.504	3	6
Office Supplies	Appliances	114.9	5	34
Furniture	Tables	1706.184	9	85
Technology	Phones	911.424	4	68
Office Supplies	Paper	15.552	3	5
Office Supplies	Binders	407.976	3	133
Office Supplies	Appliances	68.81	5	-124
Office Supplies	Binders	2.544	3	-4
Office Supplies	Storage	665.88	6	13
Office Supplies	Storage	55.5	2	10
Office Supplies	Art	8.56	2	2

5 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 16:14



Sign: _____.